How thick must the sandwich be? Thick enough to bend gravitational force lines—emerging from our brane—parallel to our brane and hold them there, so we in our brane see gravity obey an inverse square law. But no thicker, because added thickness means greater total transverse shrinkage, which may cause trouble for bulk-based adventures. (Suppose our whole universe, as seen from outside the AdS layer, were shrunk to the size of a pin head!) The required thickness turns out to be about 3 centimeters (roughly an inch), so as you travel from our brane to a confining brane, distances parallel to our brane shrink by fifteen powers of ten: a thousand trillion.

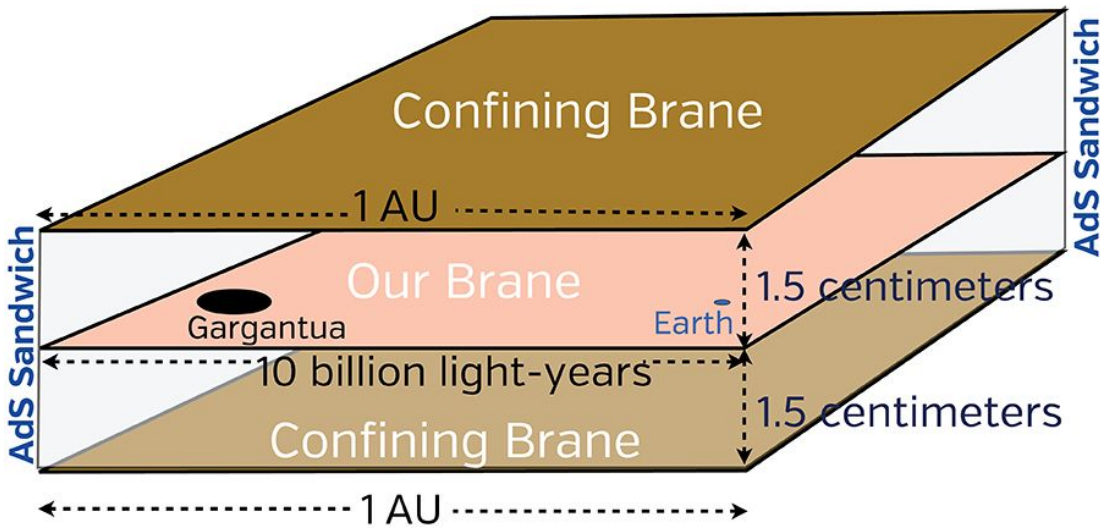


Fig. 23.7. The AdS sandwich between two confining branes. The AdS layer between the branes is lightly grayed.

In my interpretation of *Interstellar*, Gargantua is in the far reaches of the observable universe: roughly 10 billion light-years from Earth. Cooper, in the tesseract, rises through the AdS layer, from Gargantua's core into the bulk. There the distance to Earth is 10 billion light-years divided by a thousand trillion, which is about the same as the distance between the Sun and the Earth, one "astronomical unit" (1 AU; Figure